Benchmark Transmission, Environmental Stress, and Smallholder Resilience in the Cocoa Supply Chain: Evidence from Santander, Colombia

Leonardo H. Talero-Sarmiento[1] Gabriel Weber[2] Ignazio Cabras[3]
Katia Picaud-Bello[2] Stamatis Mantziaris[4]

[1] Universidad Autonoma de Bucaramanga, Colombia
[2] ESSCA School of Management, France
[3] Northumbria University, UK
[4] ELGO-DIMITRA, Greece
ltalero@unab.edu.co

Abstract

Global cocoa benchmarks shape producer exposure, but territorial resilience also depends on the environmental conditions under which benchmark shocks are absorbed. This article examines short-run transmission from the world cocoa benchmark to Colombian producer-linked prices and to a European downstream chocolate index, then extends that market analysis with contextual weather and disaster information for Santander, Colombia. Monthly AgroNet, World Bank, Eurostat, exchange-rate, oil, and NASA POWER series are harmonized for August 2021–December 2025, alongside an official departmental disaster registry covering 594 events from August 2021–July 2024. The empirical design separates a core transmission system from a nested territorial exposure layer. HAC-robust level and return models identify the benchmark pass-through mechanism, while weather anomalies, direct hazard screening, a PCA-based composite pressure index, and an exploratory event-window comparison are used as contextual resilience tools rather than as co-equal transmission series. International cocoa returns remain the dominant short-run driver of Colombian cocoa returns ($\beta \approx 0.80$, $p < 0.001$), while the European downstream index adjusts more weakly. Weather variables add little incremental explanatory power to the core equations, but they identify months in which benchmark exposure coincides with local stress. In the shorter disaster window, earthquake counts are too sparse for direct use; hydrometeorological counts provide the most defensible monthly episode marker, while the composite index summarizes broader territorial disruption. Both signals intensify around October 2022, when Colombian returns exhibit a significant mean shift, although hazard coefficients remain weak and should not be interpreted as short-run price leadership. The contribution is therefore not a disaster-led pricing model, but a resilience-oriented extension showing how benchmark transmission and territorial exposure can be analyzed jointly under short and uneven data overlap.

Keywords: cocoa price transmission; environmental stress; hydrometeorological events; smallholder vulnerability; resilience; Santander; Colombia

1 Introduction

Cocoa-producing territories are exposed to more than price risk alone. In resilience-oriented food-systems analysis, harm depends on how shocks interact with exposure, sensitivity, and adaptive capacity rather than on any single disturbance taken in isolation [21, 2, 18, 23]. For cocoa smallholders in Santander, Colombia, that interaction is visible in two connected domains: benchmark volatility arriving through international commodity markets, and localized environmental stress linked to rainfall disruption, transport damage, and other disaster-related conditions that can interfere with harvest, marketing, or household coping.

The empirical literature usually separates these domains. Cocoa and commodity-market studies explain pass-through from world prices, exchange-rate exposure, and uneven adjustment along the pricing chain [20, 10, 13, 9]. A parallel body of work documents climatic suitability, water and heat stress, and the buffering role of agroforestry and diversified production systems in cocoa landscapes [17, 11, 15]. What remains less developed is an empirical design that traces how benchmark shocks reach producer-linked prices while also incorporating defensible information on environmental stress and disaster conditions in the same territorial setting. That gap matters not only for academic interpretation, but also for smallholders, cooperatives, local traders, and territorial planners who must manage vulnerability when market instability and localized disruptions overlap.

This article addresses that gap by integrating two earlier manuscript lines into one scientific article. The first line is a supply-chain transmission analysis linking Colombian cocoa prices, world cocoa benchmarks, downstream European chocolate prices, the COP/USD exchange rate, and Brent oil prices. The second line is a disaster-resilience extension based on an official Santander event registry. The integrated design proceeds in layers. It first estimates the core market-transmission mechanism on the fullest aligned monthly window available. It then brings environmental stress into the analysis in two complementary ways: a full-window weather-stress index built from local NASA POWER anomalies, and a nested disaster extension for August 2021–July 2024 that treats registry-based hazards as territorial exposure markers rather than as a twin market-transmission system. Because the disaster data are shorter, irregular, partly zero-inflated, and partly synthetic, they enter through descriptive overlays, parsimonious robustness checks, and an exploratory event window instead of being forced into a co-equal benchmark-like specification.

The paper makes three modest but concrete contributions. First, it preserves the benchmark-transmission logic of the original manuscript while tightening the resilience framing so hazards are not confined to the abstract. Second, it evaluates how disaster information can be incorporated without overstating what the data support, showing that hydrometeorological event counts are the most defensible monthly episode marker in the nested window, while a PCA-based multi-hazard indicator remains useful only as a contextual territorial-pressure overlay. Third, it reframes the contribution in ecological-economics terms: the cocoa chain is treated as a socio-ecological system in which transmission defines exposure, while environmental stress conditions how that exposure is absorbed or amplified.

The remainder of the article is organized as follows. Section 2 develops the conceptual framing.

Section 3 describes the territorial context, data architecture, and alignment decisions. Section 4 presents the empirical design, equations, and hazard-selection logic. Section 5 reports the descriptive, transmission, and disaster-overlay findings. Section 6 interprets those results in relation to vulnerability, resilience, and governance. Section 7 concludes with limitations, transferability, and practical implications.

2 Conceptual Framing

Resilience in commodity systems is not adequately captured by prices, agronomy, or institutions taken separately. Vulnerability frameworks instead emphasize that shocks matter through their interaction with sensitivity and adaptive capacity [21, 2]. In a cocoa-producing territory, that implies that benchmark price shocks, environmental stress, and chain governance should be treated as related layers of one socio-ecological problem rather than as disconnected topics. This perspective is consistent with food-systems research that treats market, environmental, and institutional pressures as linked processes across scales [6, 18, 23].

From the market side, the key issue is transmission. Evidence from cocoa-market integration shows that stronger linkage to world prices can increase producer exposure to international volatility even when farmers do not capture a commensurate share of favorable movements [20]. More broadly, pricing-chain studies show that commodity and energy shocks often pass through asymmetrically, incompletely, or with different timing across stages of the chain [13, 9]. Those findings matter for resilience because they shift the question from whether prices are volatile to who absorbs adjustment and under what institutional conditions. In that sense, benchmark transmission is not a side issue in vulnerability analysis; it is part of the exposure mechanism itself.

From the environmental side, the relevant question is not whether every localized hazard directly sets monthly cocoa prices. Rather, it is whether climatic and disaster-related stressors help identify the context in which market shocks are experienced. Cocoa studies document sensitivity to heat and moisture stress, limits to climatic suitability, and the role of agroforestry and diversified production systems in buffering exposure [17, 11, 15]. Producer-oriented studies also emphasize weak organizations, buyer dependence, and adaptation constraints [14, 3]. Read together, this literature suggests that vulnerability in cocoa is jointly shaped by global market dependence and locally uneven buffering capacity.

This methodological distinction is consistent with the disaster-risk literature. Composite resilience and hazard-pressure metrics are often used to summarize territorial stress when no single indicator adequately captures local disruption, but such indices are normally interpreted as contextual constructs whose meaning depends on spatial scope, variable selection, and aggregation choices rather than as canonical market series [16, 7]. That logic fits the present article. The disaster layer is intended to locate periods of intensified territorial pressure and to enrich the resilience interpretation, not to mimic the stochastic properties of the benchmark price system.

The ecological-economics relevance of the paper lies in that joint interpretation. This is not presented as a new theory of resilience. The contribution is more modest and more empirical: it operationalizes a bridge between benchmark transmission and territorial stress in a setting where the data do not support strong causal claims about every hazard channel. Environmental stress therefore enters the paper in a disciplined way. Weather anomalies are used as full-window contextual indicators. Official disaster records are used in a nested window where the overlap is sufficient. Hazard claims are narrowed to what the project data can defend, especially hydrometeorological and broader multi-hazard disruption rather than unsupported general statements about all natural hazards.

The empirical design is guided by three expectations. First, world cocoa returns should transmit positively and robustly to Colombian producer-linked returns. Second, the European downstream chocolate index should adjust more slowly and less strongly than the producer-linked series, indicating uneven absorption of benchmark volatility. Third, environmental stress should be visible primarily as a contextual amplifier of exposure rather than as a stable substitute for the benchmark transmission channel.

3 Study Context and Data

The territorial focus is Santander, Colombia's leading cocoa-producing department, with particular attention to San Vicente de Chucuri as the local climate reference point. The territory combines strong cocoa specialization with exposure to rainfall variability, transport disruption, and multi-hazard events that matter for market access and household resilience. This makes Santander a plausible setting for studying how global benchmark shocks interact with local stress conditions instead of treating them as separate risk domains.

3.1 Data Sources

The integrated dataset combines four blocks. The first is the cocoa pricing chain: AgroNet weekly cacao reference prices aggregated to monthly Colombian producer-linked prices, the World Bank cocoa benchmark, and the Eurostat EU chocolate consumer price index. The second is a macro block composed of the COP/USD exchange rate and Brent oil prices, included because exchange-rate and energy movements help structure commodity-price transmission [10, 9]. The third is a local weather block from NASA POWER for San Vicente de Chucuri, including precipitation, solar radiation, and temperature variables. The fourth is an official Santander disaster registry containing 594 event records between August 2021 and July 2024, with event categories, municipalities, and damage-related variables.

Figure 1 shows the long-run coverage of the main price and macro series. The historical depth is uneven, which is why the article distinguishes clearly between long-run descriptive coverage and the much shorter aligned estimation windows.

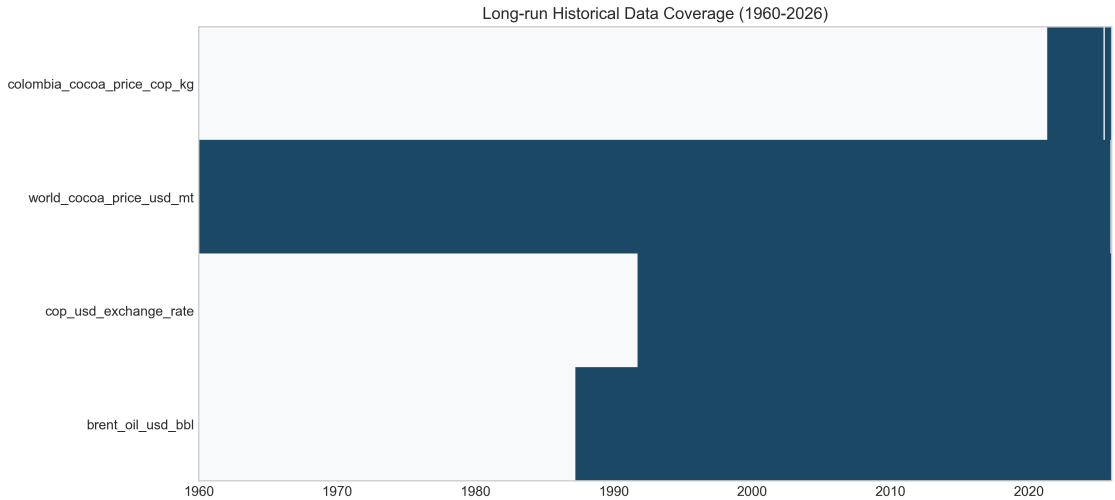

Figure 1: Long-run historical coverage of the main price and macro series used in the integrated analysis. The figure highlights the difference between long repository coverage and the much shorter aligned estimation windows imposed by the Colombian domestic price series and the disaster registry.

3.2 Variable Card and Alignment

Table 1 summarizes the principal variables. The five core series define the transmission system. Weather variables and disaster indicators provide territorial stress information rather than replacing the benchmark system, and the disaster block is interpreted as a nested contextual layer rather than as a co-equal econometric price series.

Table 1: Variable data card for the integrated analysis

Variable	Operational definition	Unit	Source	Usable window
Colombian cocoa price	Monthly average of AgroNet weekly cacao reference prices	COP/kg	AgroNet	2021-08 to 2026-03
World cocoa benchmark	Monthly international cocoa price	USD/ton	World Bank	1960-01 to 2026-02
EU chocolate index	EU27 chocolate consumer price index	Index	Eurostat	2016-12 to 2025-12
COP/USD exchange rate	Monthly average peso-dollar exchange rate	COP/USD	BanRep	1991-11 to 2026-03
Brent oil price	Europe Brent monthly average spot price	USD/barrel	EIA	1987-05 to 2026-03
NASA precipitation	Monthly mean precipitation for San Vicente de Chucuri	mm/day	NASA POWER	2000-01 to 2025-12
NASA solar radiation	Monthly mean surface solar radiation	MJ/m^2/day	NASA POWER	2000-01 to 2025-12
NASA max temperature	Monthly mean maximum air temperature	°C	NASA POWER	2000-01 to 2025-12
Hydrometeorological events	Monthly count of registry-coded hydrometeorological events	Count	Disaster registry	2021-08 to 2024-07
PCA disaster pressure	First principal component of monthly multi-hazard disruption features	Std. index	Disaster registry	2021-08 to 2024-07

Table 2 reports the aligned windows used in the article. The full merged panel contains 795 monthly observations from January 1960 to March 2026, but direct comparative estimation is constrained by the overlap of the Colombian domestic series, the Eurostat downstream series, and the disaster registry. One domestic cocoa observation in September 2025 and one solar-radiation observation in December 2025 are completed through block-specific iterative imputation for continuity-sensitive figures. Those imputations do not create longer spans of observed data and are treated as transparent continuity devices rather than substitutes for additional information [22, 19]. The disaster registry is aggregated at the departmental level, the weather series are point-based for San Vicente de Chucuri, and the cocoa price series is a broader producer-linked reference. This scale mismatch is acceptable for contextual resilience analysis, but it cautions against reading the disaster layer as a precise local price-setting series.

Table 2: Sample design and aligned estimation windows

Panel or artifact	Obs.	Date span	Role in the article
Full merged monthly panel	795	1960-01 to 2026-03	Long-run descriptive context
Core aligned levels window	53	2021-08 to 2025-12	Main levels comparisons and figures
Core aligned return window	52	2021-09 to 2025-12	Main short-run transmission models
Weather-augmented complete sample	51	2021-08 to 2025-12	Contextual weather extension
Nested disaster levels window	36	2021-08 to 2024-07	Disaster descriptives and registry aggregation
Nested disaster return window	35	2021-09 to 2024-07	Direct hazard and event-window models

4 Methodology

4.1 Transformations, Volatility, and Diagnostic Sequence

For every strictly positive price series $P_{i,t}$, the main short-run object of interest is the log return

$$r_{i,t} = \Delta \ln(P_{i,t}) = \ln(P_{i,t}) - \ln(P_{i,t-1}), \tag{1}$$

which expresses the proportional monthly shock transmitted through the chain. This transformation is appropriate both statistically and substantively. It improves comparability across variables measured in different units, reduces the dominance of level scale differences, and maps directly onto the paper's concern with shock transmission and instability rather than with raw nominal changes alone.

To characterize instability rather than average movement alone, the paper also uses a rolling volatility proxy based on the 12-month rolling standard deviation of returns:

$$\sigma_{i,t}^{(12)} = \sqrt{\frac{1}{m_t - 1} \sum_{j=0}^{m_t - 1} \left(r_{i,t-j} - \bar{r}_{i,t}^{(12)} \right)^2}, \tag{2}$$

where m_t is the number of non-missing returns available inside the rolling window up to month t. Stationarity diagnostics rely primarily on ADF tests in levels and returns, supplemented by ARCH-LM checks for volatility clustering. Given the short aligned span, these diagnostics are used to guide specification choice rather than to justify over-parameterized long-run systems [8].

4.2 Core Transmission Models

The baseline levels association for the producer-linked Colombian price is

$$\ln(P_t^{COL}) = \alpha_0 + \alpha_1 \ln(P_t^{WLD}) + \alpha_2 \ln(FX_t) + \alpha_3 \ln(OIL_t) + u_t, \tag{3}$$

where P_t^{COL} is the Colombian cocoa price, P_t^{WLD} the world cocoa benchmark, FX_t the COP/USD exchange rate, and OIL_t the Brent oil price. The short-run transmission model is

$$r_t^{COL} = \beta_0 + \beta_1 r_t^{WLD} + \beta_2 r_t^{FX} + \beta_3 r_t^{OIL} + \varepsilon_t. \tag{4}$$

An analogous pair of models is estimated for the downstream EU chocolate index. All regressions are estimated with HAC-robust standard errors to account for heteroskedasticity and serial correlation in monthly data.

The levels equations provide a compact description of long-run co-movement in the short aligned span, but the paper does not treat them as proof of fully identified long-run equilibrium. For that reason, pairwise Engle-Granger and Granger-causality diagnostics are reported only as complementary evidence and are interpreted cautiously as reduced-form diagnostics rather than as strong structural claims.

4.3 Weather-Based Contextual Stress

The weather block is used to represent local environmental stress without forcing the entire climate system into the core transmission specification. Following the project's prior variable-selection logic and local cocoa relevance, the weather-stress index is built from precipitation, surface solar radiation, and maximum temperature:

$$\text{WeatherStress}_t = \frac{1}{J} \sum_{j=1}^{J} \left| z(w_{j,t}) \right|, \tag{5}$$

where $w_{j,t}$ denotes the selected weather variable and $z(\cdot)$ its within-sample standardization. The resulting index is interpreted as a contextual anomaly score. In addition, weather-extended domestic models are estimated as a parsimonious sensitivity exercise in which lagged standardized weather covariates are added to Equation 4. Their role is diagnostic: they show whether local environmental stress materially changes the transmission estimates or mainly sharpens the vulnerability interpretation.

4.4 Contextual Disaster Exposure Layer

The disaster registry is not modeled as a second transmission system. Four features rule out that interpretation. First, the disaster window is shorter than the core return window. Second, registry-based hazard counts are sparse, irregular, and sometimes strongly zero-inflated. Third, the composite disaster indicator pools heterogeneous event, damage, and spread variables into a synthetic territorial-pressure construct. Fourth, the registry is aggregated at a different spatial scale from both the point-based weather series and the broader producer-linked price series. For these reasons, the disaster layer is used only as a contextual exposure block through descriptive overlays, a parsimonious one-at-a-time hazard screen, and an exploratory event-window comparison, rather than being forced into a large joint VAR/VECM with the benchmark system [8, 12, 1, 16, 7].

The disaster registry is handled in two stages. First, candidate direct hazard series are screened within the nested disaster return window. The screening focuses on monthly variation, the number of non-zero months, and conceptual interpretability. Isolated earthquake counts are assessed explicitly because they were the original candidate hazard series in the second manuscript version, but they are too sparse for direct time-series use. Hydrometeorological counts, geophysical counts, and total monthly event counts are then evaluated as alternative contextual signals. The objective is not to identify a co-equal price driver, but to determine whether any registry-based monthly count is dense enough to serve as a parsimonious territorial episode marker.

The selected direct hazard overlay uses standardized hydrometeorological counts H_t:

$$r_t^{COL} = \gamma_0 + \gamma_1 r_t^{WLD} + \gamma_2 r_t^{FX} + \gamma_3 r_t^{OIL} + \gamma_4 H_t + \nu_t, \tag{6}$$

while the corresponding volatility specification is

$$\sigma_t^{COL} = \delta_0 + \delta_1 \sigma_t^{WLD} + \delta_2 H_t + \upsilon_t. \tag{7}$$

These models are intentionally modest. They test whether a direct monthly hazard count adds information once the benchmark transmission channel is already controlled for. Even in this specification, H_t is interpreted as an exposure marker appended to the core model, not as a benchmark-like driver with an autonomous transmission role.

Second, a synthetic multi-hazard robustness overlay is constructed using principal component analysis on 21 eligible standardized monthly disaster features, including event frequency, municipal spread, human impact, housing damage, and infrastructure damage:

$$D_t = \sum_{k=1}^{K} w_k z(F_{k,t}). \tag{8}$$

This PCA indicator is not treated as the primary result when a direct hazard series is feasible. Instead, it functions as a robustness layer that summarizes broader multi-hazard pressure and helps identify whether the direct hazard peak is also visible in the composite registry structure. In line with the composite-indicator literature, D_t is interpreted as a synthetic territorial-pressure construct rather than as a canonical economic time series [16, 7].

4.5 Event-Based Mean-Shift Design

Because disaster effects may appear as discrete stress episodes rather than stable linear coefficients, the analysis also implements a simple intervention-style event-window comparison around the peak hazard month t^*:

$$r_t^{COL} = \theta_0 + \theta_1 \mathbf{1}(t \geq t^*) + e_t, \tag{9}$$

evaluated in symmetric six-month windows before and after the identified peak. In practice, the main inference is based on Welch, Levene, and Kolmogorov-Smirnov comparisons across the pre- and post-peak windows. The peak month is selected from the hazard screen itself, which makes the event window exploratory and data-driven rather than an externally imposed quasi-experiment. This event-based design is used because the short disaster overlap is more compatible with transparent intervention logic than with an overextended dynamic hazard system.

5 Results

5.1 Descriptive Structure and Statistical Properties

The aligned sample already shows why the paper separates the benchmark system from the environmental block. Table 3 reports the levels and dispersion of the main series in the aligned window. Colombian and world cocoa prices are far more volatile than the EU downstream chocolate index, while the local weather series display distinct seasonal and anomaly structures rather than simple co-trending price behavior.

Table 3: Descriptive statistics for the core system and selected weather variables in the aligned window

Series	Mean	Std. dev.	Median	Min	Max
Colombian cocoa price	17,334.75	9,168.37	12,536.69	7,926.08	34,395.33
World cocoa benchmark	4,943.44	2,660.61	3,630.14	2,239.13	10,745.11
EU chocolate index	128.83	19.73	125.20	104.25	164.99
COP/USD exchange rate	4,147.16	302.33	4,047.41	3,773.38	4,926.66
Brent oil price	82.61	13.41	80.92	62.54	122.71
NASA precipitation	5.73	3.59	5.08	0.03	14.67
NASA max temperature	31.93	3.04	30.79	28.67	39.15
NASA solar radiation	18.24	1.19	18.18	15.68	20.62

Table 4 shows that the core price series behave as non-stationary processes in levels over the aligned window, whereas return transformations are much more stable. Several weather series reject the ADF unit-root null already in levels, which reinforces the decision to treat environmental stress as a contextual block rather than to force the whole system into one long-run specification.

Table 4: Selected statistical properties from the aligned panel

Series	Mean level	Annualized vol.	ADF level p	ADF return p	ARCH-LM p
Colombian cocoa price	17,334.75	0.376	0.643	0.000	0.768
World cocoa benchmark	4,943.44	0.367	0.648	0.000	0.013
EU chocolate index	128.83	0.030	0.984	0.013	0.005
COP/USD exchange rate	4,147.16	0.110	0.525	0.000	0.903
Brent oil price	82.61	0.258	0.460	0.081	0.083
NASA precipitation	5.73	3.106	0.000	0.006	0.070
NASA max temperature	31.93	0.152	0.018	0.003	0.438
NASA solar radiation	18.25	0.283	0.000	0.000	0.434

Figure 2 illustrates the proportional co-movement that defines the core transmission problem. Colombian and world cocoa prices move closely through the 2024–2025 boom and correction, whereas the EU downstream index follows a much smoother path.

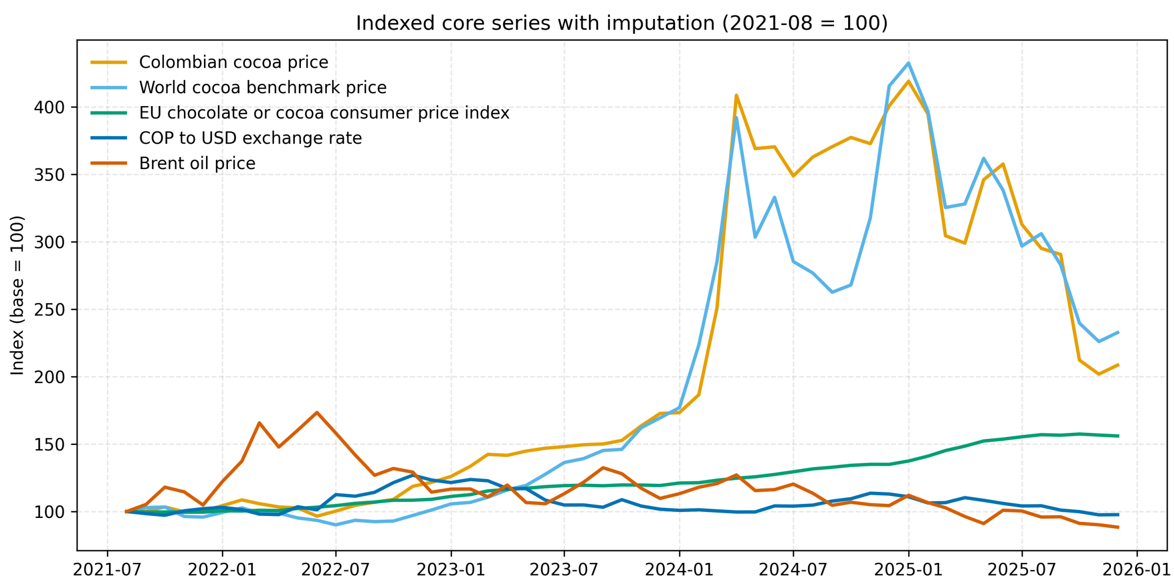

Figure 2: Indexed core series in the aligned monthly window (August 2021 = 100). The close co-movement between the Colombian and world cocoa series contrasts with the smoother adjustment of the downstream EU chocolate index.

5.2 Supply-Chain Transmission

Table 5 reports the main transmission elasticities. The central result is stable across the earlier manuscript versions and remains the strongest result in the integrated article: world cocoa returns are the dominant short-run driver of Colombian producer-linked returns. In the core return model, the world benchmark coefficient is 0.796 ($p < 0.001$). The corresponding levels association is also close to one. By contrast, the downstream EU chocolate index has a weaker and slower link to the world benchmark. Its levels coefficient is positive, but the return specification has very low fit and a small negative short-run coefficient.

Table 5: Domestic and downstream price transmission results

Model component	Coefficient	Std. error	p-value	Adj. R^2
Producer-linked domestic price				
$\ln(P_t^{COL}) \leftarrow \ln(P_t^{WLD})$	0.990	0.037	< 0.001	0.965
$\ln(P_t^{COL}) \leftarrow \ln(FX_t)$	0.902	0.147	< 0.001	0.965
$\Delta \ln(P_t^{COL}) \leftarrow \Delta \ln(P_t^{WLD})$	0.796	0.180	< 0.001	0.581
$\Delta \ln(P_t^{COL}) \leftarrow \Delta \ln(FX_t)$	0.218	0.204	0.287	0.581
Downstream EU chocolate index				
$\ln(P_t^{EU}) \leftarrow \ln(P_t^{WLD})$	0.296	0.126	0.018	0.727
$\Delta \ln(P_t^{EU}) \leftarrow \Delta \ln(P_t^{WLD})$	-0.023	0.010	0.023	-0.007

This pattern supports an uneven-adjustment interpretation. Producer-linked prices absorb benchmark volatility rapidly, while the downstream consumer layer appears more buffered. Pairwise Granger results, retained in the supplementary section, support predictive precedence from world to Colombian cocoa at lag 3 but also show reverse predictive links in the short sample. The appropriate conclusion is therefore reduced-form rather than structural: benchmark transmission is the clearest observable exposure mechanism, but the short aligned span does not warrant stronger claims about a fully identified dynamic chain.

5.3 Environmental Stress, Hazards, and Resilience Overlay

Environmental stress is present beyond the disaster registry. The weather-extended domestic models retained from the first manuscript version improve fit only modestly relative to the core transmission equations (increase in adjusted R^2 of 0.0017 in levels and 0.0139 in returns), and most individual weather coefficients remain statistically weak. That finding is substantively useful because it supports a contextual interpretation of local environmental stress rather than a claim that weather replaces benchmark transmission as the main short-run price mechanism. Figure 3 summarizes this full-window contextual layer through the weather-stress score and the exploratory exposure indices.

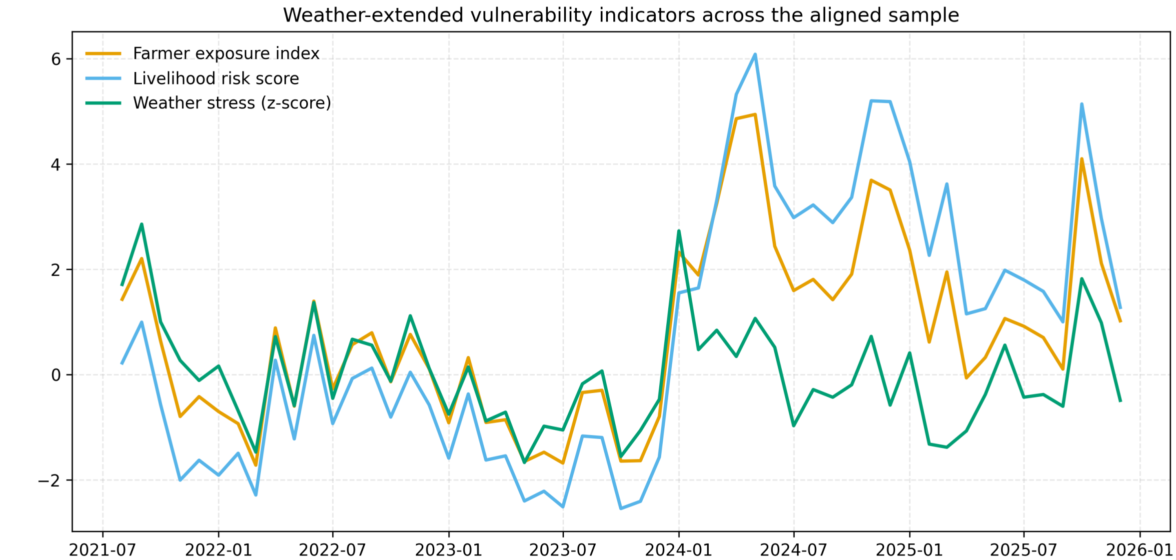

Figure 3: Weather-extended contextual indicators across the aligned sample. The figure summarizes local weather anomalies together with benchmark-driven exposure and volatility proxies, showing when environmental stress and market instability accumulate in the same months.

The nested disaster window sharpens that interpretation. Figure 4 shows the monthly event totals and hazard-domain mix for Santander. Hydrometeorological and geophysical events dominate the coded event structure, whereas isolated earthquakes are infrequent.

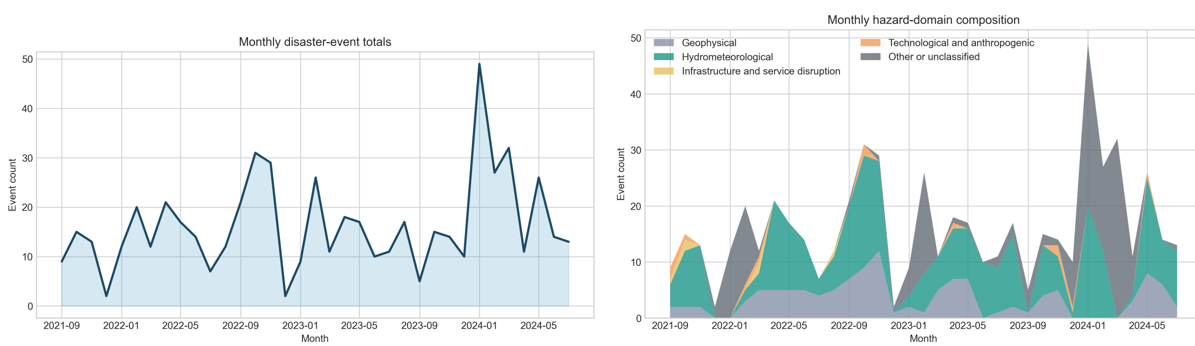

Figure 4: Disaster event dynamics and hazard-domain composition in Santander, August 2021–July 2024. Monthly totals reveal recurrent pressure rather than isolated single-shock episodes, and the hazard mix is dominated by hydrometeorological and geophysical events.

Figure 5 aligns the world and Colombian cocoa returns with the hydrometeorological count and the composite territorial-pressure index in the nested window. The contrast in signal shape is central to the paper's revised methodology. The benchmark series move as continuous market returns, whereas the disaster series behave as episodic and uneven territorial signals. Their value lies in identifying months of intensified local pressure, not in behaving like a second benchmark.

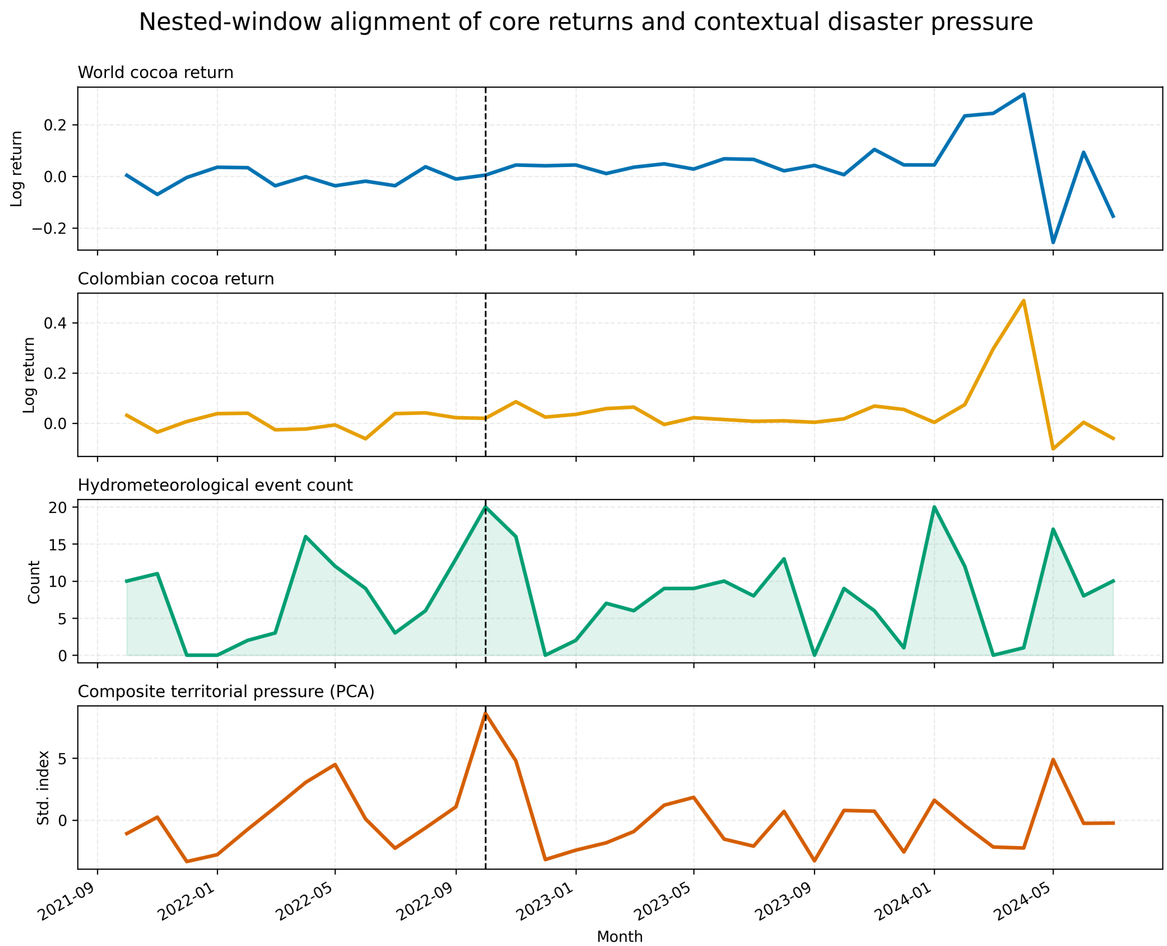

Figure 5: Nested return-window alignment of world and Colombian cocoa returns with hydrometeorological counts and the composite territorial-pressure index. The vertical line marks October 2022, the exploratory peak month identified from the hazard screen. The contrast in signal shape underscores that the disaster layer is episodic and contextual rather than benchmark-like.

Table 6 formalizes the screening of possible contextual hazard overlays in the 35-month nested return window. Earthquake counts are too sparse to support direct integration. Hydrometeorological events are observed in 30 of the 35 months and peak in October 2022, making them the most viable monthly territorial episode marker among the candidate counts.

Table 6: Screening of contextual hazard-overlay candidates in the nested return window

Signal	Months	Non-zero months	Zero share	Peak month	Peak value
Earthquake events	35	4	0.886	2022-09	1
Geophysical events	35	28	0.200	2022-11	12
Hydrometeorological events	35	30	0.143	2022-10	20
Total recorded events	35	35	0.000	2024-01	49

Table 7 then compares the nested-window hazard overlays. These estimates should be read as parsimonious contextual checks rather than as a horse race among co-equal transmission drivers. The hydrometeorological series outperforms the other direct count alternatives in both the return and volatility extensions. Its coefficient is negative and only marginally significant ($p = 0.083$ in returns; $p = 0.074$ in volatility), so it should not be interpreted as a dominant short-run price setter. Even so, it is more informative than geophysical counts, total monthly events, or the continuous PCA index when the objective is to select one direct territorial episode marker.

Table 7: Contextual hazard-overlay comparison in the nested return window

Signal	Type	Return adj. R^2	Return coef.	Return p	Vol. adj. R^2	Vol. coef.	Vol. p
Hydrometeorological events	Direct	0.653	-0.015	0.083	0.935	-0.003	0.074
Geophysical events	Direct	0.633	0.004	0.613	0.932	0.002	0.286
Total recorded events	Direct	0.632	-0.002	0.823	0.932	-0.002	0.140
PCA disaster pressure	Composite	0.633	-0.004	0.562	0.932	-0.002	0.264

The synthetic robustness overlay nevertheless adds useful information. The first principal component explains 33.1% of the standardized variance across 21 eligible monthly disaster features, and its peak occurs in the same month as the hydrometeorological series. Figure 6 shows that the composite indicator and Colombian cocoa returns both intensify around October 2022. The continuous PCA coefficient is weak, but the composite series helps identify a multi-hazard stress episode that is broader than one hazard family alone. The relevant interpretation is contextual: the index summarizes territorial disruption across several dimensions, but it is not treated as a conventional transmission series.

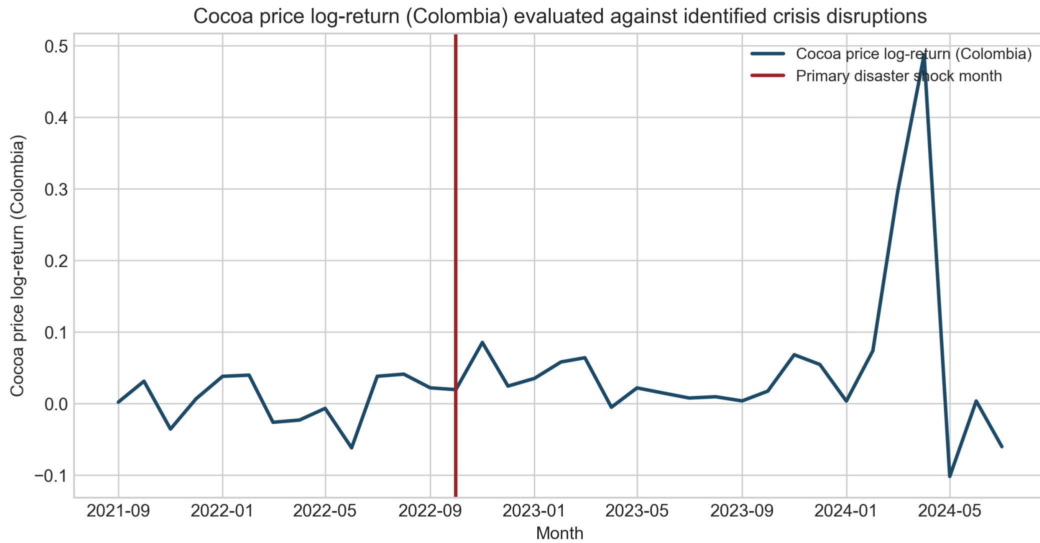

Figure 6: Composite multi-hazard pressure and cocoa return instability in the nested disaster window. The peak stress episode occurs in October 2022, which also corresponds to the peak month of the direct hydrometeorological event series. The figure is interpreted as an episode-screening tool rather than as evidence of benchmark-like price leadership.

Table 8 reports the event-window comparison around that peak. Mean domestic returns rise from 0.002 in the six months before October 2022 to 0.048 in the six months after, with a Welch p-value of 0.043. Variance and distribution shifts are weaker. The implication is that the disaster overlay is more visible as a discrete stress episode than as a stable linear marginal effect. Because October 2022 was selected from the same nested hazard data used in the screen, this comparison should be read as exploratory episode stratification rather than as exogenous shock identification.

Table 8: Exploratory event-window comparison around the October 2022 peak contextual-pressure month

Metric	Before peak	After peak	Statistic	p-value
Mean return shift (Welch test)	0.002	0.048	-2.378	0.043
Variance shift (Levene test)	0.002	0.001	1.338	0.274
Distribution shift (KS test)	0.008	0.047	0.500	0.474

6 Discussion

The integrated article shows that resilience in the cocoa chain cannot be reduced to a climatic narrative or to a market narrative alone. The strongest and most stable result remains benchmark transmission: Colombian producer-linked prices move closely with world cocoa prices, and that short-run link is economically large. In vulnerability terms, this means that Santander smallholders operate as price takers in a global system whose benchmark shocks are transmitted quickly to the producer-linked layer. That observation grounds the resilience argument in an observable mechanism of exposure rather than in a purely rhetorical claim [21, 2, 20].

The downstream results reinforce the same point from another angle. The European chocolate index shows a positive levels association with the benchmark but far weaker return adjustment. This does not prove a full downstream insulation mechanism, and the aligned sample is too short to support strong chain-wide structural claims. Even so, the contrast is consistent with an uneven-adjustment interpretation in which producer-linked prices absorb benchmark volatility more directly than the downstream consumer layer. In resilience terms, the question is not simply whether volatility exists, but where it is absorbed and whose livelihoods are exposed when shocks arrive.

Environmental stress does not overturn that transmission logic, but it changes how the transmission results should be interpreted. The weather block improves model fit only slightly, which argues against recasting the article as a weather-dominant price model. At the same time, the full-window stress indicators show that benchmark-driven exposure often occurs under locally unfavorable environmental conditions. The nested disaster extension sharpens the point further. Among the direct hazard candidates, hydrometeorological counts are the most defensible monthly territorial episode marker, and they peak in the same month as the multi-hazard PCA indicator. Their negative coefficients in the nested overlay models are suggestive rather than definitive, but they are consistent with the possibility that local disruption weakens the local realization of otherwise benchmark-driven price movements through marketing, logistics, or timing constraints. The October 2022 mean shift reinforces the interpretation that territorial shocks matter more as crisis windows than as stable month-to-month price setters.

This is where the ecological-economics framing becomes useful. The paper's contribution is not to claim that disasters directly determine cocoa prices in a short monthly sample. It is to show that vulnerability emerges from the coupling of market transmission and territorial stress. Benchmark transmission defines the exposure channel; local weather anomalies, hydrometeorological disruption, and multi-hazard event pressure condition the sensitivity and buffering capacity of the local system. The article therefore extends a standard market-integration study into a socio-ecological interpretation without overstating causal inference.

Several methodological cautions are central to that interpretation. The PCA disaster indicator is not directly comparable to world cocoa, exchange-rate, or oil prices. It is a synthetic territorial-pressure construct built from heterogeneous event, damage, and spread variables, of the kind often used in disaster-risk assessment to summarize contextual exposure rather than market dynamics [16, 7]. Likewise, the monthly hydrometeorological count is an irregular registry-based count process, not a canonical continuous price series. Weak coefficients, weak Granger evidence, or isolated pairwise correlations therefore do not justify presenting the disaster layer as a dominant driver of the market system. The shorter disaster window is substantively important here: with only 35 return observations, sparse event counts, and heterogeneous indicator construction, a large multivariate transmission model would imply more comparability than the data can support [8, 12].

The event-window evidence should be read with the same discipline. October 2022 was selected from the same nested hazard data used to identify the most viable direct overlay, so the resulting comparison is exploratory rather than strictly exogenous. There is also a scale mismatch between the departmental disaster registry, the point-based weather series, and the broader producer-linked price system. Those limits do not eliminate the relevance of territorial disruption, but they do narrow the claim: the disaster layer helps identify periods in which benchmark pass-through is experienced under adverse local conditions; it does not establish a disaster-led short-run price process.

That narrower claim is still substantively meaningful. Resilience scholarship emphasizes that local coping capacity, infrastructure, and governance condition how shocks are absorbed even when those shocks originate elsewhere [5, 4]. The present results fit that pattern. A price shock and a territorial disruption do not need to be identical in form to interact. Benchmark transmission can remain the main exposure mechanism while territorial disruption amplifies how that exposure is experienced by smallholders, cooperatives, and local traders.

The weak short-run exchange-rate coefficient in the return model also deserves a cautious institutional interpretation. One plausible explanation is that sector coordination, contracting practices, or price-setting arrangements dampen immediate currency pass-through even when world cocoa benchmarks remain dominant. This interpretation is not directly observed in the present data and should therefore remain conditional. Even so, it points toward a practical resilience question: institutional arrangements may not eliminate exposure to world benchmarks, but they may influence how quickly secondary shocks are transmitted to producers.

Several limitations remain. The core aligned return sample contains 52 monthly observations, and the nested disaster return window contains only 35. That is enough for transparent reduced-form analysis, hazard screening, and event-window comparisons, but not for strong claims from large multivariate cointegration systems. The Colombian domestic price series is a national reference price rather than a farm-gate panel, and the weather block is location-based rather than household-specific. The disaster registry is rich enough for monthly aggregation, but still short for finely identified dynamic hazard models. These limits argue for empirical modesty, not for abandoning the resilience question.

Even with those constraints, the integrated findings yield practical lessons. Resilience-enhancing arrangements in cocoa-producing territories are unlikely to come from adaptation planning alone if benchmark transmission remains the dominant exposure mechanism. Risk-sharing instruments, cooperative coordination, storage and marketing support, contract duration, diversification, and local early-warning systems are more plausibly complementary than competing responses. For territorial governance, the main implication is that market shocks and environmental disruption should be monitored together rather than treated as separate policy files.

7 Conclusion

This article integrates two earlier manuscript versions into one publication-oriented analysis of cocoa price transmission and territorial resilience in Santander, Colombia. The main conclusion is straightforward: international cocoa benchmarks remain the dominant short-run driver of Colombian producer-linked prices, but the vulnerability implications of that transmission are better understood once environmental stress is incorporated into the empirical design. Weather anomalies matter primarily as contextual stress. In the shorter disaster window, hydrometeorological event counts provide the most defensible monthly territorial episode marker, while the PCA-based multi-hazard index remains useful as a robustness overlay and event-screening device. The October 2022 peak shared by both signals is associated with a significant mean shift in domestic cocoa returns, but the disaster layer is best understood as contextual territorial exposure rather than as a co-equal transmission series.

The contribution is therefore one of empirical integration and methodological discipline rather than exaggerated novelty. The paper shows how a benchmark-transmission framework can be connected to resilience analysis without claiming more than the data support. Its transferability is strongest in territories and commodities where producers are strongly tied to global benchmarks but also face recurrent environmental disruption under short and uneven data overlap. Future research should test whether longer local price histories, farm-gate panels, independently identified hazard episodes, or more detailed contracting data can identify which institutional arrangements actually buffer joint market-and-environmental shocks. Until then, the most defensible practical lesson is that producer resilience requires attention to both price transmission and territorial exposure, including coordination mechanisms, diversification, early-warning systems, risk-sharing instruments, and adaptation planning that recognize their interaction.

Supplementary Display Items

The following supplementary figures and tables preserve the diagnostic evidence base developed across the first two manuscript versions. Items that became redundant in the main text were retained here rather than discarded.

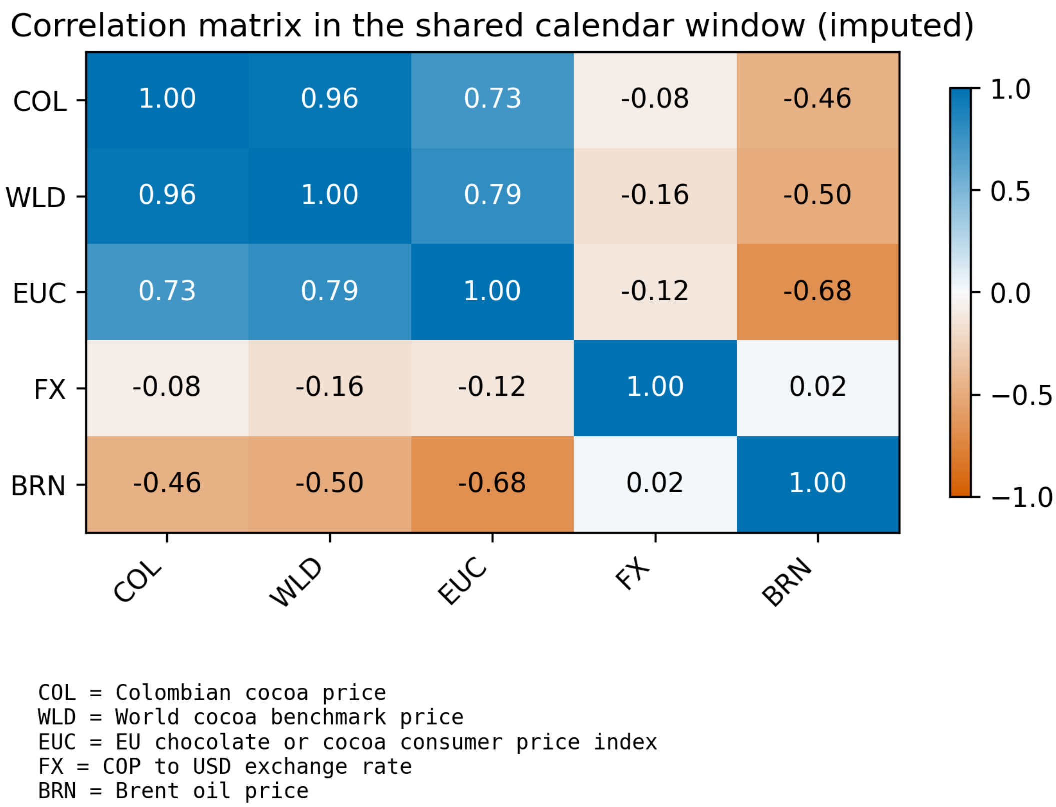

Figure 7: Core-system level correlation heatmap retained from the first manuscript version. The display preserves the original descriptive evidence on the dependence structure of the benchmark, domestic, downstream, exchange-rate, and oil series.

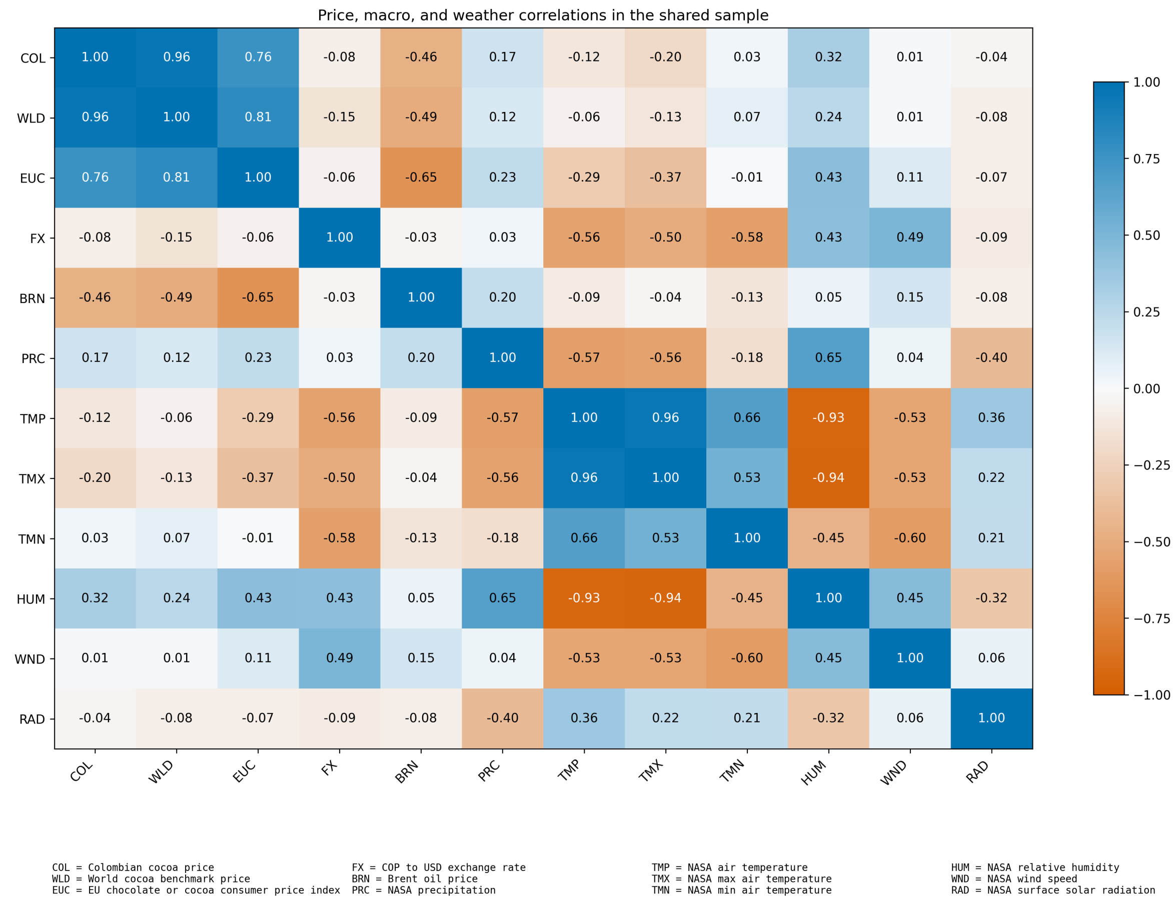

Figure 8: Climate-core correlation matrix retained from the first manuscript version. The figure preserves the original bridge between the price system and the local weather block without requiring a large unstable climate-augmented VAR.

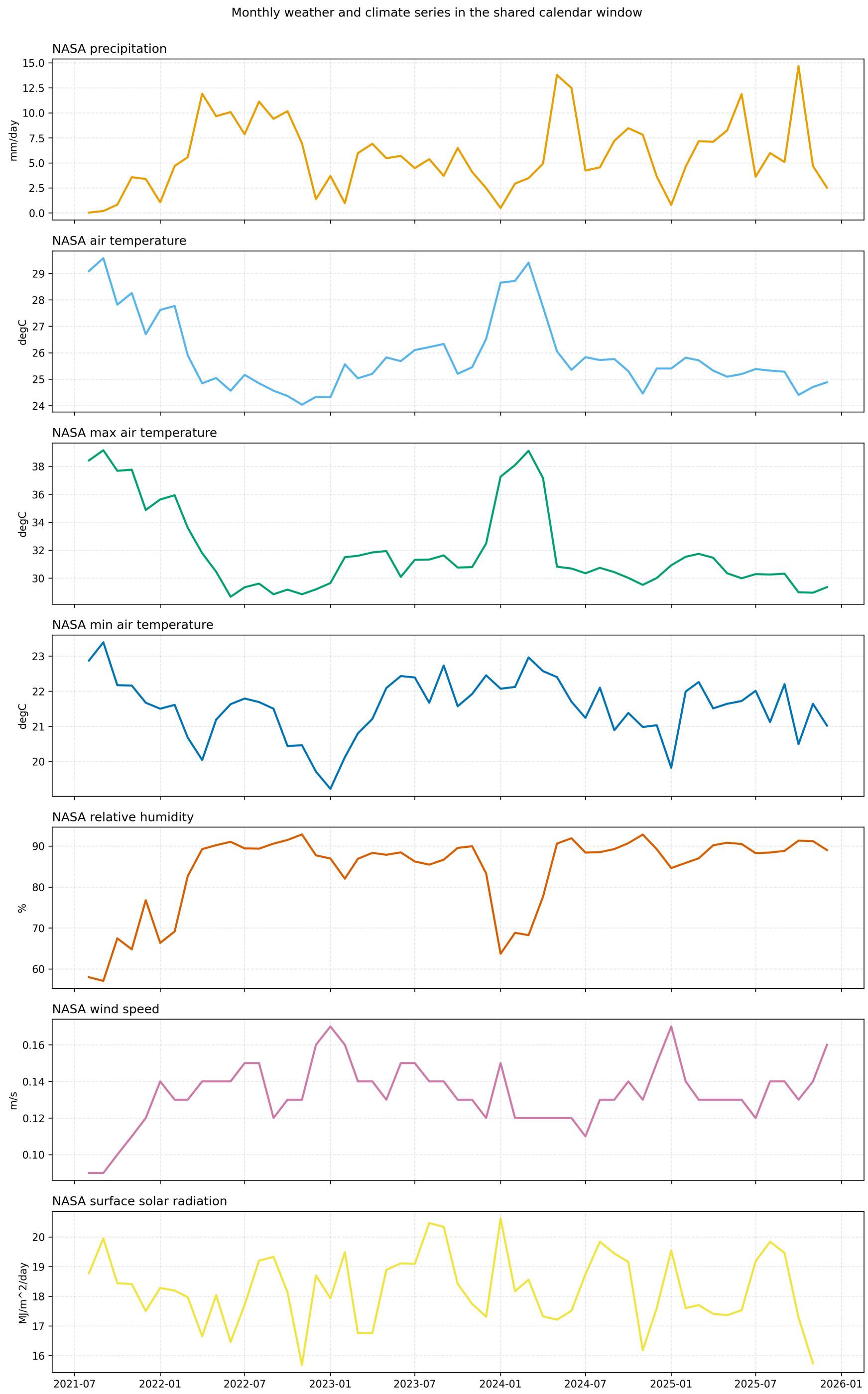

Figure 9: Climate series panels for San Vicente de Chucuri retained from the first manuscript version. These panels preserve the underlying environmental context summarized more compactly by the main-text weather-stress figure.

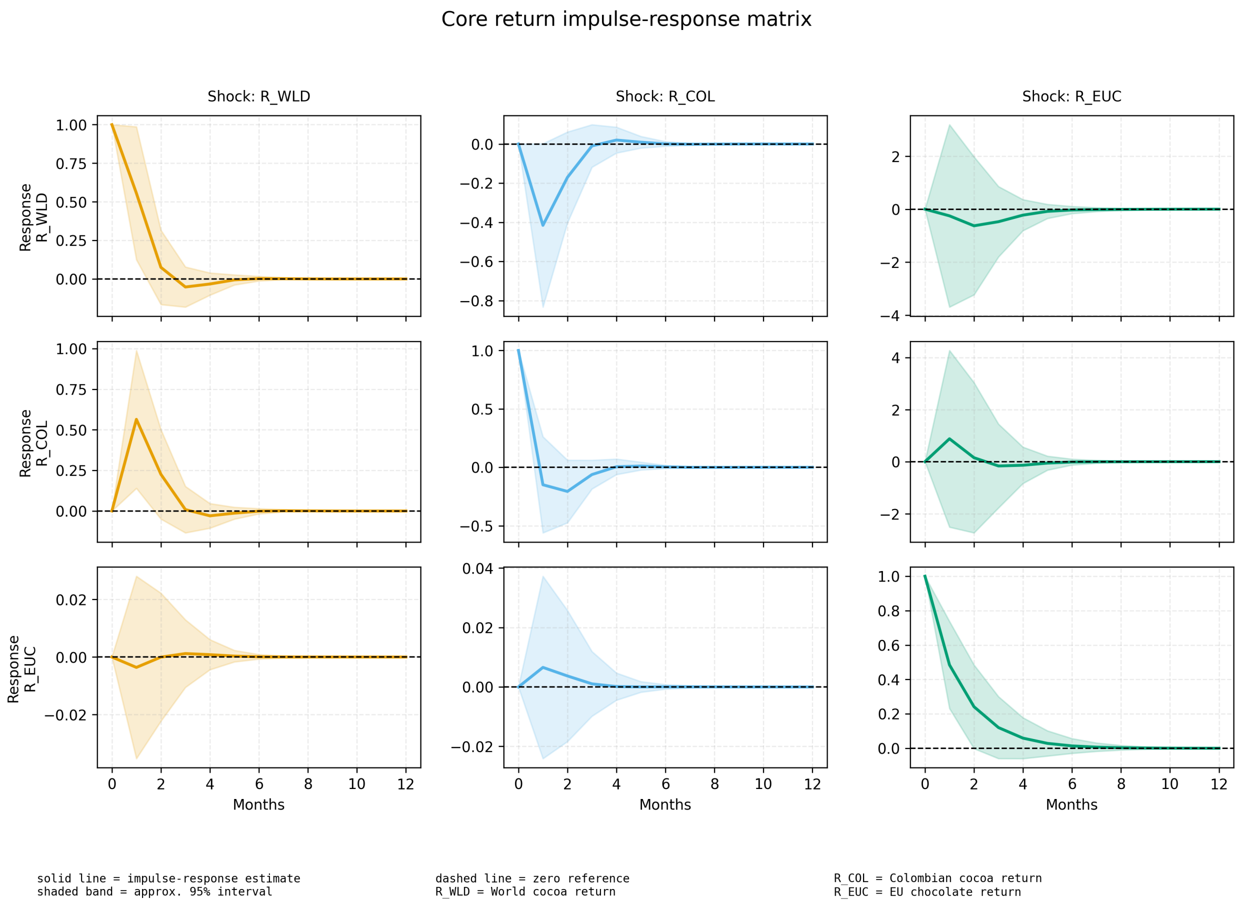

Figure 10: Compact return-VAR impulse-response matrix retained from the first manuscript version. Because the aligned sample is short, the article treats this figure as supplementary reduced-form evidence rather than as a headline causal result.

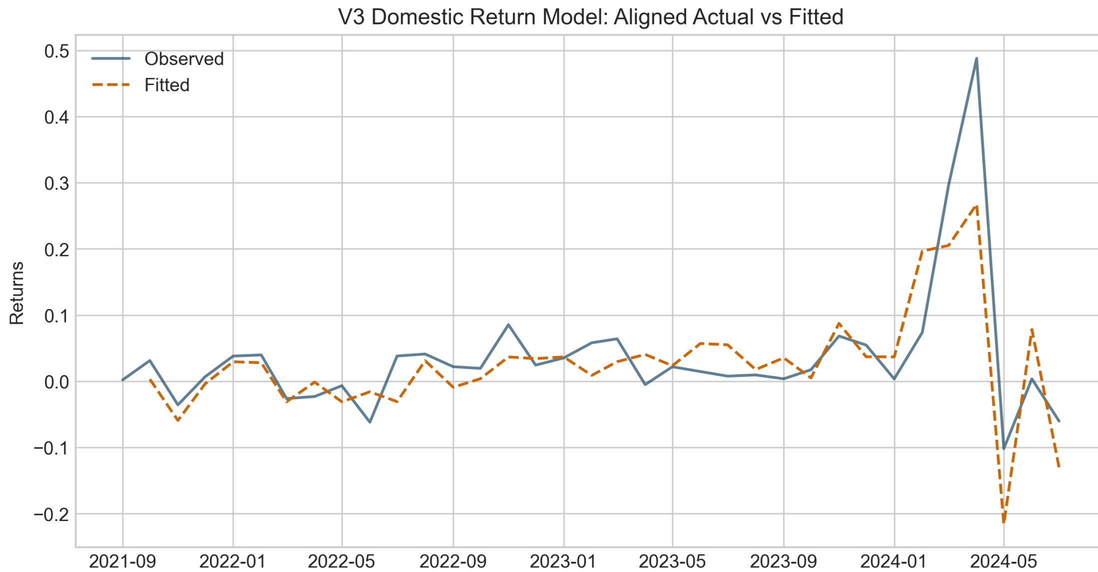

Figure 11: Actual-versus-fitted domestic return model from the disaster-extension manuscript. The figure is retained as a supplementary model-diagnostic display after the main article shifted the emphasis toward hazard screening and event-window interpretation.

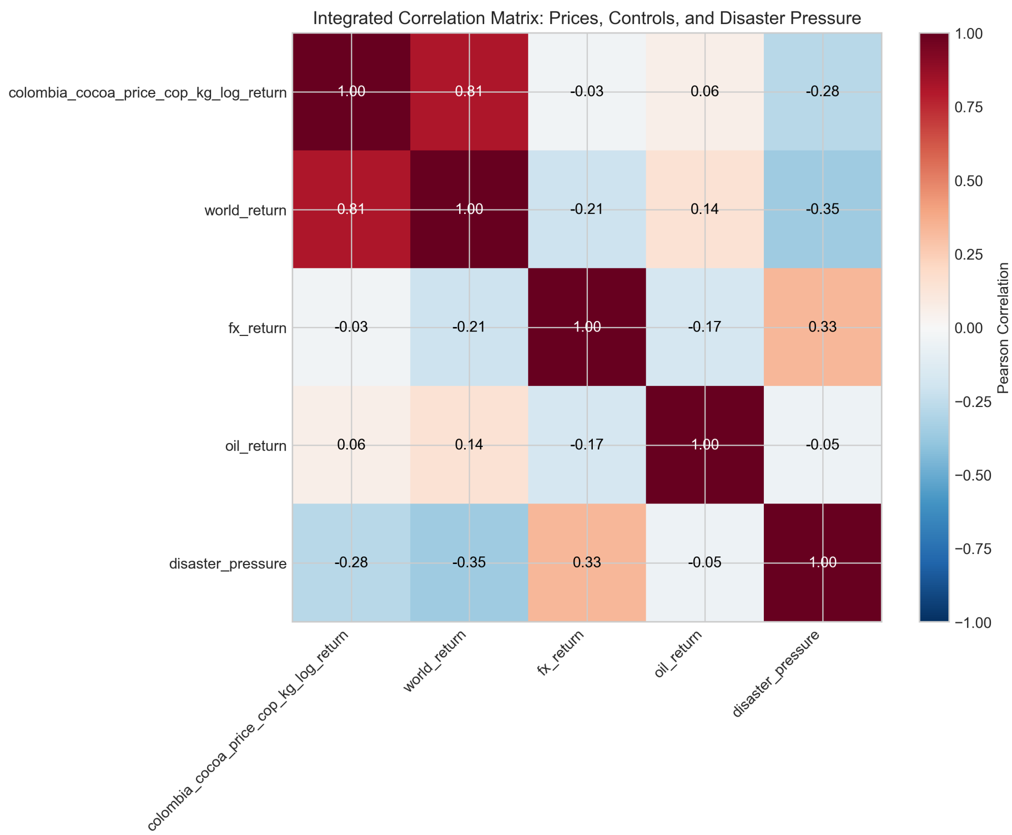

Figure 12: Integrated correlation matrix from the disaster-extension manuscript. The figure is retained to preserve the original pairwise descriptive view of returns, macro controls, and composite disaster pressure.

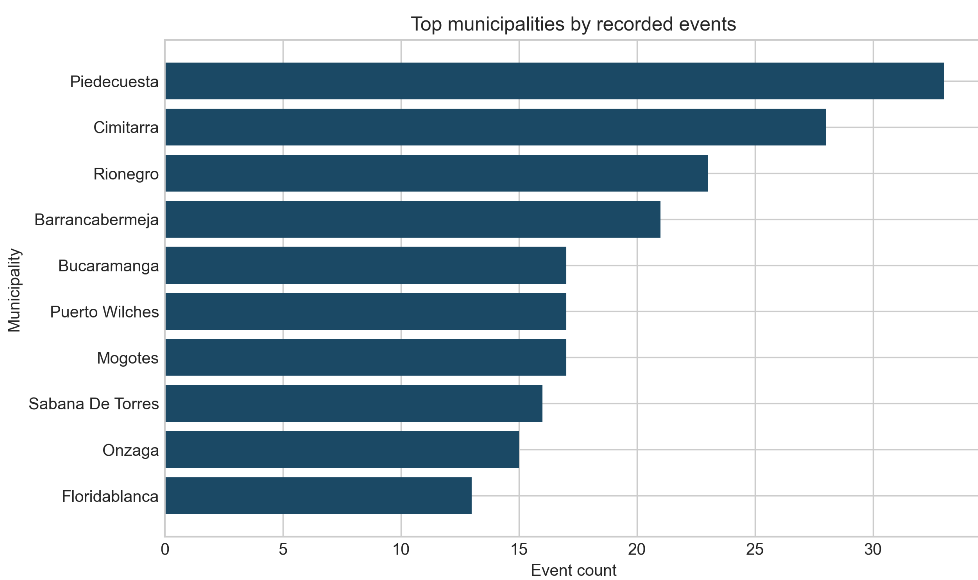

Figure 13: Top municipalities by recorded events retained from the disaster-extension manuscript. The figure is kept as supplementary territorial detail rather than as a main-text result.

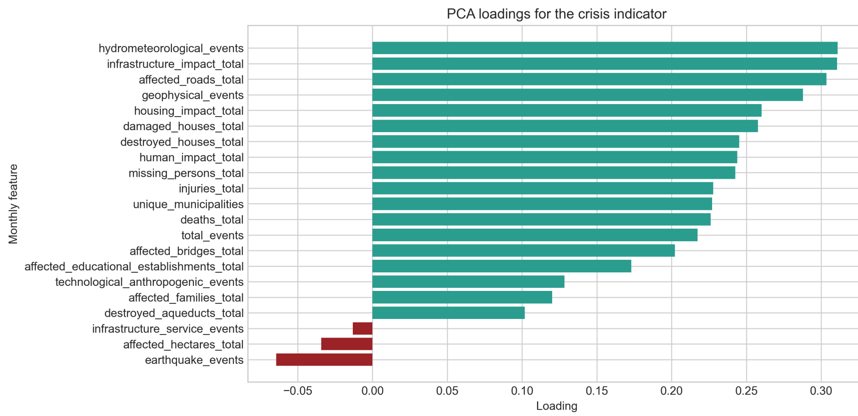

Figure 14: Loadings for the first principal component of the multi-hazard disaster-pressure indicator. The figure is retained to document the composition of the synthetic robustness overlay used in the integrated article.

Table 9: Lag-specific pairwise Granger-causality p-values in the aligned core return system

Direction	Lag 1 p	Lag 2 p	Lag 3 p	Best lag (p)
World cocoa → Colombian cocoa	0.050	0.075	0.023	3 (0.023)
Colombian cocoa → World cocoa	0.013	0.076	0.027	1 (0.013)
Colombian cocoa → EU chocolate	0.979	0.801	0.871	2 (0.801)
EU chocolate → Colombian cocoa	0.692	0.781	0.641	3 (0.641)
World cocoa → EU chocolate	0.678	0.597	0.364	3 (0.364)
EU chocolate → World cocoa	0.866	0.191	0.192	2 (0.191)

Table 10: Weather-extended domestic models retained from the first manuscript version

Model component	Coefficient	Std. error	p-value	Adj. R^2
$\ln(P_t^{COL}) \leftarrow \ln(P_t^{WLD})$	0.977	0.037	< 0.001	0.966
$\ln(P_t^{COL}) \leftarrow$ solar radiation$_t$ (z)	0.028	0.016	0.075	0.966
$\ln(P_t^{COL}) \leftarrow$ precipitation$_t$ (z)	0.024	0.020	0.247	0.966
$\ln(P_t^{COL}) \leftarrow$ max temperature$_t$ (z)	-0.009	0.019	0.627	0.966
$\Delta \ln(P_t^{COL}) \leftarrow \Delta \ln(P_t^{WLD})$	0.773	0.157	< 0.001	0.595
$\Delta \ln(P_t^{COL}) \leftarrow$ max temperature$_{t-1}$ (z)	0.023	0.015	0.119	0.595
$\Delta \ln(P_t^{COL}) \leftarrow$ precipitation$_{t-1}$ (z)	0.003	0.010	0.725	0.595
$\Delta \ln(P_t^{COL}) \leftarrow$ solar radiation$_{t-1}$ (z)	-0.008	0.008	0.330	0.595
Domestic rolling volatility $\leftarrow$ world rolling volatility	0.954	0.051	< 0.001	0.917
Domestic rolling volatility $\leftarrow$ weather stress$_{t-1}$	0.013	0.012	0.253	0.917

Table 11: Exploratory vulnerability indicators retained from the first manuscript version

Indicator	Construction	Mean	Std. dev.
Weather stress index	Mean absolute standardized anomaly across precipitation, solar radiation, and maximum temperature	0.798	0.356
Weather stress (z-score)	Standardized weather-stress index; positive values indicate above-average combined anomaly	0.000	1.010
Farmer exposure index	$z(\sigma_t^{COL}) + z(\widehat{m}_t) + \text{WeatherStress}_t$	0.758	1.691
Livelihood risk score	Farmer exposure index plus standardized world-cocoa volatility	0.758	2.429

Table 12: Disaster-indicator Granger-causality matrix retained from the disaster-extension manuscript

Source	Target	Lag 1 p	Lag 2 p	Lag 3 p	Lag 4 p
Disaster indicator	Colombian cocoa return	0.876	0.987	0.983	0.967
Disaster indicator	World cocoa return	0.654	0.958	0.961	0.775
Disaster indicator	FX return	0.443	0.762	0.830	0.545
Disaster indicator	Oil return	0.383	0.575	0.677	0.627

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